

50. Let $\frac{x}{1776} = \frac{111}{x}$. Then, x^2
 $= 111 \times 1776 = 111 \times 111 \times 16$
 $\Rightarrow x = \sqrt{(111)^2 \times (4)^2} = 111 \times 4 = 444.$
51. Let $\frac{4\frac{1}{2}}{x} = \frac{x}{32}$. Then, $x^2 = 32 \times \frac{9}{2} = 144 \Leftrightarrow x = \sqrt{144} = 12.$
52. Let $\frac{x}{\sqrt{128}} = \frac{\sqrt{162}}{x}$. Then, $x^2 = \sqrt{128 \times 162} = \sqrt{64 \times 2 \times 18 \times 9}$
 $= \sqrt{8^2 \times 6^2 \times 3^2} = 8 \times 6 \times 3 = 144.$
 $\therefore x = \sqrt{144} = 12.$
53. $\sqrt{x + \frac{x}{y}} = x \sqrt{\frac{x}{y}} \Rightarrow x + \frac{x}{y} = x^2 \cdot \frac{x}{y} \Rightarrow \frac{xy + x}{y} = \frac{x^3}{y}$
 $\Rightarrow xy + x = x^3$
 $\Rightarrow y + 1 = x^2$
 $\Rightarrow y = x^2 - 1.$
54. $n^2 = (25)^{64} \times (64)^{25} = (5^2)^{64} \times (2^6)^{25}$
 $= 5^{128} \times 2^{150} = 5^{128} \times 2^{128} \times 2^{22}$
 $\Rightarrow n = 5^{64} \times 2^{64} \times 2^{11} = (5 \times 2)^{64} \times 2^{11} = 10^{64} \times 2048.$
 $\therefore$ Sum of digits of $n = 2 + 0 + 4 + 8 = 14.$
55. $\frac{0.13}{p^2} = 13$
 $\Leftrightarrow p^2 = \frac{0.13}{13} = \frac{1}{100}$
 $\Leftrightarrow p = \sqrt{\frac{1}{100}} = \frac{1}{10} = 0.1.$
56. Let the required number be x . Then, $\frac{x}{\sqrt{0.25}} = 25$
 $\Leftrightarrow \frac{x}{0.5} = 25$
 $\Leftrightarrow x = 25 \times 0.5 = 12.5.$
57. $\sqrt{3^n} = 729 = 3^6 \Leftrightarrow (\sqrt{3^n})^2 = (3^6)^2 \Leftrightarrow 3^n = 3^{12} \Leftrightarrow n = 12.$
58. $\sqrt{18 \times 14 \times x} = 84 \Leftrightarrow 18 \times 14 \times x = 84 \times 84$
 $\Leftrightarrow x = \frac{84 \times 84}{18 \times 14} = 28.$
59. Let $28\sqrt{x} + 1426 = 3 \times 718.$
 Then, $28\sqrt{x} = 2154 - 1426 \Leftrightarrow 28\sqrt{x} = 728 \Leftrightarrow \sqrt{x} = 26$
 $\Leftrightarrow x = (26)^2 = 676.$
60. Let $\sqrt{\frac{x}{169}} = \frac{54}{39}$. Then, $\frac{\sqrt{x}}{13} = \frac{54}{39} \Leftrightarrow \sqrt{x} = \left(\frac{54}{39} \times 13\right) = 18$
 $\Leftrightarrow x = (18)^2 = 324.$
61. $\frac{\sqrt{x}}{\sqrt{441}} = 0.02 \Leftrightarrow \frac{\sqrt{x}}{21} = 0.02$
 $\Leftrightarrow \sqrt{x} = 0.02 \times 21 = 0.42 \Leftrightarrow x$
 $= (0.42)^2 = 0.1764.$

62. Let $\sqrt{\frac{.0196}{x}} = 0.2$. Then, $\frac{.0196}{x} = 0.04$
 $\Leftrightarrow x = \frac{.0196}{.04} = \frac{1.96}{4} = .49.$
63. Let $\sqrt{0.0169 \times x} = 1.3$. Then, $0.0169x = (1.3)^2 = 1.69$
 $\Rightarrow x = \frac{1.69}{0.0169} = 100.$
64. $37 + \sqrt{.0615 + x} = 37.25 \Leftrightarrow \sqrt{.0615 + x} = 0.25$
 $\Leftrightarrow .0615 + x = (0.25)^2 = 0.0625$
 $\Leftrightarrow x = .001 = \frac{1}{10^3} = 10^{-3}.$
65. $\sqrt{(x-1)(y+2)} = 7 \Rightarrow (x-1)(y+2) = (7)^2 \Rightarrow (x-1) = 7$
 and $(y+2) = 7 \Rightarrow x = 8$ and $y = 5.$
66. $\frac{\sqrt{a}}{\sqrt{b}} = \frac{.004 \times .4}{\sqrt{.04 \times .4}} \Rightarrow \frac{a}{b} = \frac{.004 \times .4 \times .004 \times .4}{.04 \times .4} = \frac{.0000064}{.04}$
 $\Rightarrow \frac{a}{b} = \frac{.00064}{4} = .00016 = \frac{16}{10^5} = 16 \times 10^{-5}.$
67. Let the number be x . Then,
 $\frac{3}{5}x^2 = 126.15 \Leftrightarrow x^2 = \left(126.15 \times \frac{5}{3}\right) = 210.25$
 $\Leftrightarrow x = \sqrt{210.25} = 14.5.$
68. $\sqrt{\frac{0.361}{0.00169}} = \sqrt{\frac{0.36100}{0.00169}} = \sqrt{\frac{36100}{169}} = \frac{190}{13}.$
69. $\sqrt{\frac{48.4}{0.289}} = \sqrt{\frac{48.400}{0.289}} = \sqrt{\frac{48400}{289}} = \frac{220}{17} = 12\frac{16}{17}.$
70. $\sqrt{1 + \frac{x}{169}} = \frac{14}{13} \Rightarrow 1 + \frac{x}{169} = \frac{196}{169}$
 $\Rightarrow \frac{x}{169} = \left(\frac{196}{169} - 1\right) = \frac{27}{169} \Rightarrow x = 27.$
71. $\sqrt{1 + \frac{55}{729}} = 1 + \frac{x}{27} \Rightarrow \sqrt{\frac{784}{729}} = \frac{27+x}{27}$
 $\Rightarrow \frac{28}{27} = \frac{27+x}{27} \Rightarrow 27+x = 28 \Rightarrow x = 1.$
72. $\sqrt{\frac{4}{3}} - \sqrt{\frac{3}{4}} = \frac{\sqrt{4}}{\sqrt{3}} - \frac{\sqrt{3}}{\sqrt{4}} = \frac{\sqrt{4} \times \sqrt{4} - \sqrt{3} \times \sqrt{3}}{\sqrt{12}} = \frac{4-3}{2\sqrt{3}} = \frac{1}{2\sqrt{3}}.$
- 73.
- | | |
|------|-------------------|
| 1 | 2.00 00 00 (1.414 |
| | 1 |
| 24 | 100 |
| | 96 |
| 281 | 400 |
| | 281 |
| 2824 | 11900 |
| | 11296 |
- $\therefore \sqrt{2} = 1.414$
74. $2\sqrt{27} - \sqrt{75} + \sqrt{12} = 2\sqrt{9 \times 3} - \sqrt{25 \times 3} + \sqrt{4 \times 3}$
 $= 6\sqrt{3} - 5\sqrt{3} + 2\sqrt{3} = 3\sqrt{3}.$